Likith Kumar

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable full-stack, cloud-native, and distributed applications across financial services and digital payments. Strong in Java, Spring Boot, Python, ReactJS, Node.js, AWS, Kafka, Kubernetes, and CI/CD, with hands-on experience delivering GenAI, LLM, RAG, and machine learning solutions. Known for improving system performance, reliability, automation, and customer experiences while taking ownership of production-ready software from development through deployment.

SKILLS

- **Programming Languages:** Java 8/11/17, Python, C++17, JDBC, Servlets, JSP, JavaScript (ES6+), TypeScript, SQL, PL/SQL, HTML5, CSS3, Unix Shell Scripting
- **Backend & Microservices:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful Web Services, Microservices Architecture, Distributed Systems, API Integration, Asynchronous Processing
- **Security & Service Architecture:** OAuth 2.0, JWT Authentication, Eureka Service Discovery, Spring Cloud Config, API Gateway
- **Frontend Development:** ReactJS, Angular, HTML5, CSS3, JSP, Servlets, Responsive UI Development
- **AI/ML & Generative AI:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, OpenAI APIs, Anthropic Claude, Amazon Bedrock, Model Context Protocol (MCP), Vector Embeddings, Prompt Engineering, GitHub Copilot
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Predictive Analytics, ML Pipelines, Classification, Feature Engineering, Model Evaluation, Inference Workflows
- **Messaging & Cloud:** Apache Kafka, Kafka Streams, JMS, AWS, AWS Lambda, EC2, S3, RDS, IAM, CloudWatch, SQS, VPC, Azure
- **DevOps & Application Servers:** Docker, Kubernetes, Jenkins, CI/CD, Maven, Gradle, Apache Tomcat, IBM WebSphere, Git, GitHub, Linux/Unix
- **Databases & Data Engineering:** PostgreSQL, MySQL, Oracle, MongoDB, DB2, SQL Server, SQL Optimization, Database Indexing
- **Libraries & SDKs**: STL, Boost, C++ REST SDK (Casablanca), Apache Spark, TensorFlow, PyTorch
- **Testing & Engineering:** JUnit, Mockito, Cucumber BDD, Integration Testing, Regression Testing, Performance Testing, JMeter, Debugging, Root-Cause Analysis, MTTR Reduction, Splunk, Log4j, Agile, Scrum, SDLC, JIRA

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03)
- Create Image Captioning Models with Google Cloud

PROFESSIONAL EXPERIENCE

Software Engineer **06/2025 – Present**

PayPal, *USA*

- Architected distributed payment services using Java 17, Spring Boot, REST APIs, and Apache Kafka to decouple transaction and merchant workflows into independently scalable components, increasing service throughput by 35% and reducing p95 latency by 28% across payment engineering and merchant-platform workloads.
- Delivered full-stack transaction and merchant experiences with ReactJS, TypeScript, Node.js, and Express.js, integrating reusable frontend components with secure backend APIs to simplify customer, merchant, and support workflows and accelerate delivery of production-facing capabilities.
- Built a Generative AI and RAG platform with Python, OpenAI APIs, vector embeddings, and prompt engineering to retrieve payment documentation and operational knowledge from enterprise sources, reducing repetitive investigation effort by 32% and improving resolution workflows for engineering and support teams.
- Orchestrated Agentic AI workflows with Anthropic Claude, LangGraph, and Model Context Protocol (MCP) to connect LLM reasoning with governed application services, enabling AI-assisted support and developer workflows while enforcing controlled access across production use cases.
- Implemented centralized observability with Prometheus, Grafana, and AWS monitoring services across distributed payment applications and data stores, improving real-time production visibility and accelerating identification of performance, availability, and dependency-related issues.
- Operationalized predictive analytics and ML inference pipelines with Python, PyTorch, and Scikit-learn using AWS Lambda for scalable execution, supporting transaction-pattern analysis and operational-exception detection across risk and platform workflows.
- Strengthened production security and reliability with OAuth 2.0, JWT, AWS IAM, VPC, S3, EC2, CloudWatch, Docker, and Kubernetes, improving operational resilience and contributing to a 45% reduction in MTTR for production incidents.
- Streamlined CI/CD and engineering quality controls with Jenkins, Maven, Git, GitHub, JUnit, Mockito, and GitHub Copilot, automating build, test, and deployment checks to improve release consistency, reduce manual validation, and support faster delivery without weakening engineering governance.

Software Engineer **12/2023 – 05/2024**

PNC, *USA*

- Engineered financial-services microservices with Java 17, Spring Boot, Spring Security, and RESTful Web Services to modernize account and transaction workflows, improving average service response time by 30% for digital banking and operations stakeholders.

- Automated AWS infrastructure provisioning for database and analytics workloads using Terraform, standardizing environment configuration and reducing manual setup effort across financial-services development and deployment environments.
- Secured distributed application services with OAuth 2.0, JWT authentication, API Gateway, and Spring Security, establishing controlled access to financial workflows and sensitive data for customer-facing and internal systems.
- Connected ReactJS, Angular, TypeScript, and JavaScript interfaces with backend APIs to support responsive financial workflows, giving business and service teams clearer visibility into account and transaction activity.
- Automated application delivery with Maven, Jenkins, Git, GitHub, Docker, and Kubernetes, incorporating JUnit and Mockito into the development lifecycle to improve release consistency and testing discipline within an Agile/Scrum environment.

Software Developer **01/2021 – 07/2023**

Groww, *India*

- Designed Java and Spring Boot microservices for investment, portfolio, and account workflows to establish independently deployable services, increasing deployment frequency by 35% for product engineering teams.
- Implemented service security with Spring Security, OAuth 2.0, and JWT to protect investor accounts and API endpoints, providing controlled access to portfolio and customer information.
- Established event-driven processing with Apache Kafka, Kafka Streams, and asynchronous workflows to handle investment orders and notification events, supporting more than 5,000 events per minute across customer-facing services.
- Modernized web experiences with ReactJS, Angular, JavaScript, TypeScript, JSP, and Servlets, connecting modern interfaces with backend services while maintaining compatibility with legacy application components.
- Optimized high-traffic investment workflows with Redis caching, SQL tuning, and database indexing across PostgreSQL, MySQL, and Oracle, reducing average API response time from 200 ms to 60 ms.
- Containerized backend workloads with Docker and Kubernetes and standardized application delivery through Maven, Jenkins, Git, and GitHub, reducing deployment effort by 50% for engineering and release teams.
- Validated application quality through JUnit, Mockito, Cucumber BDD, integration testing, and regression testing, while contributing to Agile/Scrum ceremonies, sprint planning, code reviews, SDLC activities, and JIRA-based delivery.

Education

Master of Science in Computer Science **2025**
University of Alabama at Birmingham, *United States*

Bachelor of Technology **2023**
Graphic Era University, *India*